

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Password Visibility Toggle for Login and Student Add Forms

Field	Details
Change Request ID	CR-016
Change Request Title	Password Visibility Toggle for Login and Student Add Forms
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

The system required a clearer and more user-friendly way to enter passwords. Users needed the ability to temporarily view a password while entering it, reducing typing mistakes while keeping passwords masked by default. The implemented change covers the Login form and the Add Student form.

1.2 Existing Behaviour

Before implementation:

- The Login form used a masked password field without a visibility toggle.
- The Add Student form contained a password field for new student accounts.
- The Add Student form did not provide a user-controlled password visibility toggle.
- Password validation was required for newly created students.
- Existing authentication and student creation APIs were already available.

1.3 Requested / Implemented Change

The frontend was enhanced to provide:

- Show / hide password functionality in the Login form.
- Show / hide password functionality in the Add Student form.
- Masked password fields by default.
- Eye / eye-slash visual indicators.
- Accessible aria-label and aria-pressed attributes for visibility controls.
- Non-submitting toggle buttons using type="button".
- Password input spacing so the icon does not overlap entered text.
- Preservation of existing Login and Add Student submission flows.
- Field-level validation state handling in the Add Student form.

2. Implementation

2.1 Implementation Summary

CR-016 was implemented as a frontend enhancement affecting the Login form and the Add Student form.

LoginFormSection.jsx uses a React showPassword state to switch the password input between password and text types. StudentFormModal.jsx uses a separate showPassword state for the Add Student password field and includes a matching visibility toggle.

During the Add Student form implementation, validation state management was also completed by defining fieldErrors and touched React states. These states support inline validation messages and prevent runtime errors when the modal is opened or fields are edited. Existing student creation, editing, authentication, and API flows were preserved.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/LoginFormSection.jsx	Modified	Implemented login password visibility state, toggle button, dynamic password type, and accessible eye / eye-slash icons.
frontend/src/styles/login.css	Modified	Implemented login password wrapper, toggle positioning, input spacing, hover/focus/disabled states, and responsive styling.
frontend/src/components/StudentFormModal.jsx	Modified	Implemented Add Student password visibility toggle, password validation state, field error handling, touched state, and modal reset behaviour.
Student form styling	Modified / Verified	Supports the password wrapper/toggle presentation and existing student modal layout.
Backend / API / Database	No change	Existing endpoints and data structures were preserved.

2.3 Technical Changes

Login Password Visibility

- showPassword is initialized as false.
- The Login password input switches between type="password" and type="text".
- The toggle changes the icon and accessibility attributes according to the current state.
- The toggle is a button with type="button" and therefore does not submit the form.

Add Student Password Visibility

- StudentFormModal.jsx maintains showPassword state for the Add Student password.
- The password field is displayed only when creating a new student.
- The input switches between masked and visible text.
- An eye / eye-slash SVG toggle provides visual feedback.
- The state is reset when the form is opened for a new operation or closed.

Add Student Validation State

- fieldErrors stores validation messages for individual fields.
- touched stores whether a field has been interacted with.
- Blur events validate the relevant field and update its error state.
- Changing a field clears its active field error while the user corrects the value.

- Submit validation continues to enforce registration, name, email, phone, and password rules.
- The missing `fieldErrors` and `touched` state declarations were added to prevent the modal from failing at runtime.

Password Validation

- New student passwords are required.
- Passwords must contain at least 8 characters.
- Passwords must include an uppercase letter.
- Passwords must include a lowercase letter.
- Passwords must include a number.
- Passwords must include a special character.

Security and Data Preservation

- Passwords remain masked by default.
- The visibility toggle changes presentation only; it does not change the password value.
- No password is intentionally stored in browser storage by these changes.
- Existing authentication and student API requests were preserved.
- No database schema or backend authentication change was introduced.

2.4 Implementation Logic

1. Open the Login or Add Student form.
2. Initialize the relevant `showPassword` state as `false`.
3. Display the password using `type="password"`.
4. When the eye button is clicked, invert the visibility state.
5. Display `type="text"` and the eye-slash indicator when visible.
6. Return to `type="password"` and the eye indicator when hidden.
7. Keep the toggle button separate from form submission.
8. For Add Student, validate the password and other fields before submitting.
9. Use the existing API request for authentication or student creation.
10. Reset form and visibility state when the relevant form is closed or reopened.

2.5 Error and Data Handling

The Login form retains its existing authentication error handling. The Add Student form retains its existing API error handling for unauthorized and forbidden requests. Field-level errors are displayed using `fieldErrors` and `touched` state. When validation fails on submission, relevant errors are populated and displayed. The visibility toggle itself does not alter or submit the password value.

3. Testing & Verification

3.1 Testing Approach

The changes were manually verified in the development environment. Testing covered Login password visibility, Add Student password visibility, validation, UI behaviour, form submission, regression of existing operations, and the runtime issue that occurred when validation state declarations were missing.

Environment: Development

Browser: Google Chrome

Test Role: Administrator / authenticated user

Testing Types: Functional, UI, validation, accessibility, regression, error handling, and CRUD verification

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Login form	Password is masked by default	Masked password displayed	Pass
TC-02	Click Login password toggle	Password becomes visible	Password displayed as text	Pass
TC-03	Click Login toggle again	Password becomes masked	Password returned to masked state	Pass
TC-04	Verify Login icon	Eye / eye-slash reflects state	Icon changes correctly	Pass
TC-05	Toggle Login password	Form is not submitted	No unintended submission	Pass
TC-06	Open Add Student form	Modal opens without runtime error	Add Student modal opens correctly	Pass
TC-07	Add Student password default	Password is masked	Masked password displayed	Pass
TC-08	Click Add Student toggle	Password becomes visible	Password displayed as text	Pass
TC-09	Click Add Student toggle again	Password becomes masked	Password returned to masked state	Pass
TC-10	Verify Add Student icon	Icon reflects visibility state	Icon changes correctly	Pass
TC-11	Blur invalid Add Student field	Relevant validation message appears	Field error displayed	Pass
TC-12	Correct invalid Add Student field	Field error clears while editing	Error clears correctly	Pass
TC-13	Submit invalid new-student password	Password validation blocks submission	Validation message displayed	Pass
TC-14	Submit valid Add Student data	Student creation request proceeds	Existing API submission preserved	Pass
TC-15	Edit existing student	Existing edit flow remains functional	Edit operation preserved	Pass
TC-16	Login with existing credentials	Existing authentication continues to work	Login flow preserved	Pass
TC-17	Close/reopen Add Student modal	Form and visibility state reset	Fresh form displayed	Pass
TC-18	Responsive UI verification	Toggle remains positioned correctly	Password UI remains usable	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	18
Passed	18
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Login users can show and hide the password.	Login visibility test	Met
AC-002	Add Student users can show and hide the new-student password.	Add Student visibility test	Met
AC-003	Passwords are masked by default.	Initial state verification	Met
AC-004	Toggle controls do not submit the surrounding form.	Button behaviour test	Met

AC-005	Visibility icons and accessibility attributes reflect the current state.	UI/accessibility verification	Met
AC-006	Add Student validation displays appropriate password and field errors.	Validation testing	Met
AC-007	Add Student modal opens without runtime errors.	Modal opening test	Met
AC-008	Valid Add Student data continues to use the existing student creation API.	CRUD/API regression test	Met
AC-009	Existing Login authentication continues to work.	Login regression test	Met
AC-010	Existing Edit Student functionality continues to work.	CRUD regression test	Met
AC-011	Password visibility changes do not modify the actual password value.	Functional verification	Met
AC-012	Existing backend/API/database functionality is preserved.	Regression verification	Met

5. Evidence

A maximum of two relevant screenshots should be attached to the final submitted report.

Evidence 01 – Login Password Visibility Toggle/ Related: AC-001, AC-003, AC-004, AC-005

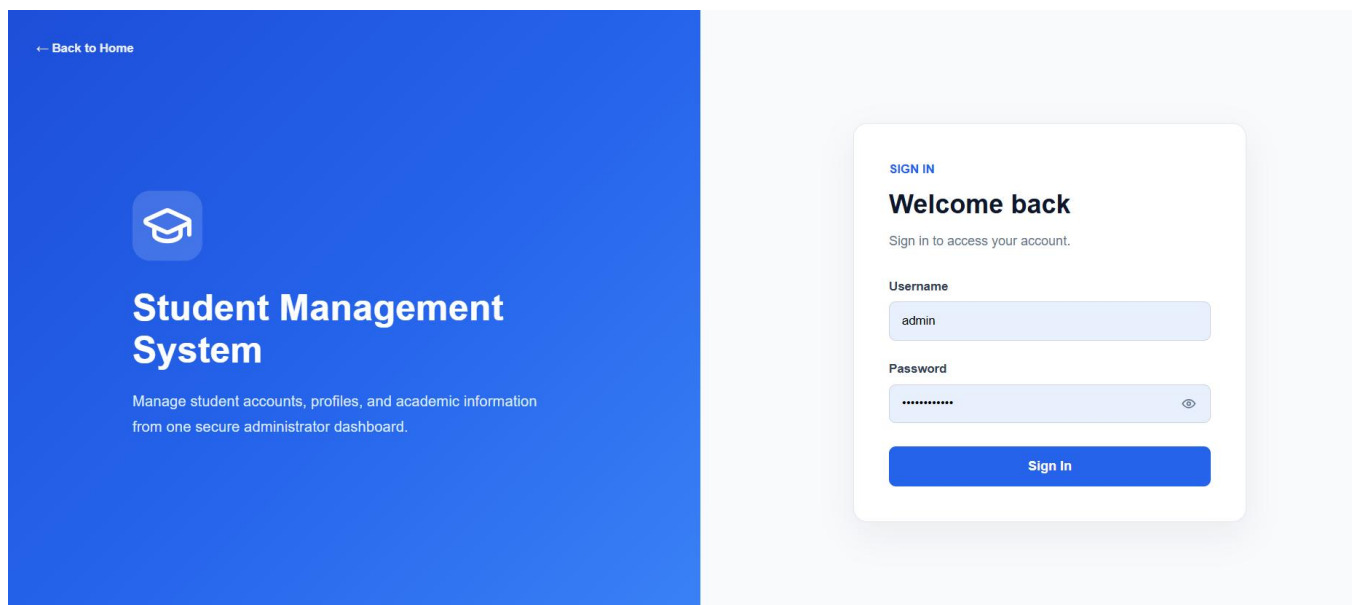


Figure 1: Login page showing the password visibility toggle in masked and visible states.

Evidence 02 – Add Student Password Toggle and Validation/ Related: AC-002, AC-006, AC-007, AC-008

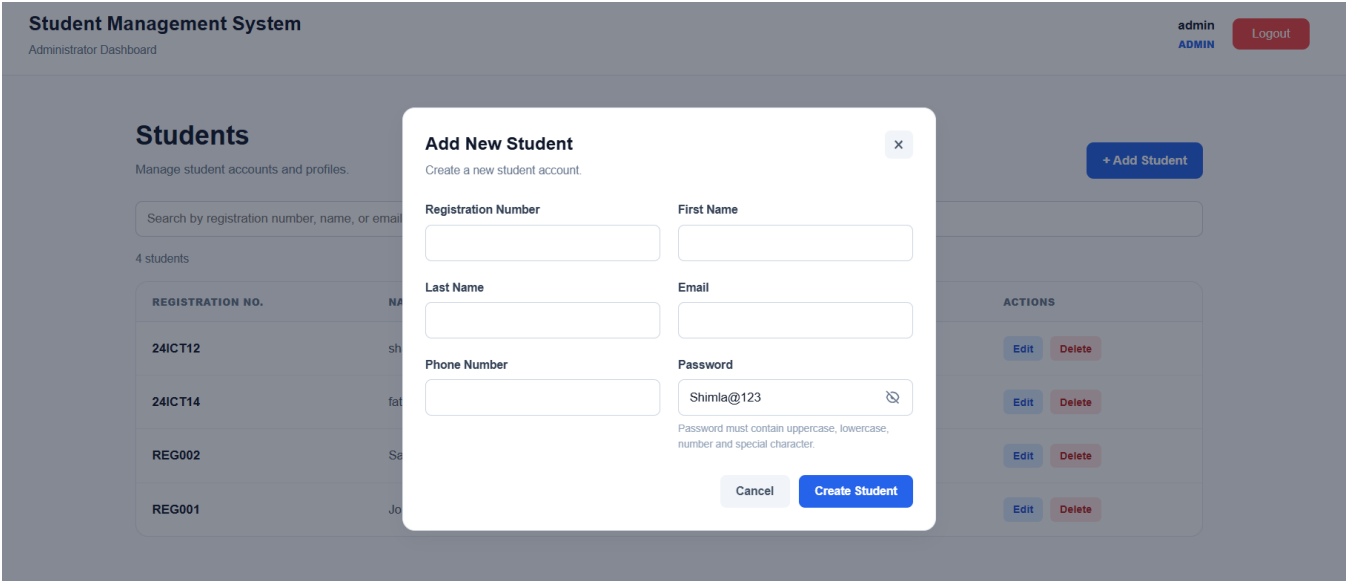


Figure 2: Add Student form showing password visibility toggle and validation behaviour.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Login Functionality	Verified / Preserved
Add Student Functionality	Verified / Preserved
Existing CRUD	Verified / Preserved
Backend / API Changes	None
Database Changes	None
Final Status	Accepted

6.1 Final Summary

CR-016 was implemented as a password usability enhancement covering both the Login module and the Add Student form. The Login form now provides a show/hide password toggle, while the Add Student form provides the same functionality for newly created student accounts. Both controls default to masked passwords and provide visual and accessible state feedback.

The Add Student implementation also includes the required fieldErrors and touched validation states, ensuring that field-level validation operates correctly and the modal opens without the runtime error that previously resulted in a blank white UI. Password validation remains enforced for new students, and existing student creation, editing, authentication, authorization, and API behaviour were preserved.

6.2 Issues / Limitations

No significant issues identified after implementation and verification. The password visibility feature is a client-side presentation enhancement; it does not change the server-side authentication or password storage mechanism.

Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Date	2026/10/08
Signature	

~ END OF REPORT ~